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Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTROTECHNOLOGY
NQF LEVEL 2

(12040212)

17 February 2026 (Y-paper)
13:00–16:00

Calculators may be used.

This question paper consists of 8 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
ELECTROTECHNOLOGY
NQF LEVEL 2
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
 2. Read all the questions carefully.
 3. Number the answers according to the numbering system used in this question paper.
 4. Use only a black or blue pen.
 5. Write neatly and legibly.
-

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QUESTION 1



- 1.1 Choose ONE term from the list below for each of the following descriptions and write it next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

atom; Faraday's law of induction; diode; Lenz's law; element;
Fleming's left-hand rule; compound; transistor; molecule

- 1.1.1 The smallest particles of a compound having all the properties of that compound
- 1.1.2 A semiconductor device that allows current to flow in one direction
- 1.1.3 State that the induced voltage in a coil is proportional to the rate of change of magnetic flux
- 1.1.4 A substance made of only one type of atom that cannot be broken down
- 1.1.5 A combination of two or more elements

(5 × 1)

(5)

- 1.2 Study FIGURE 1 below and answer the questions that follow:

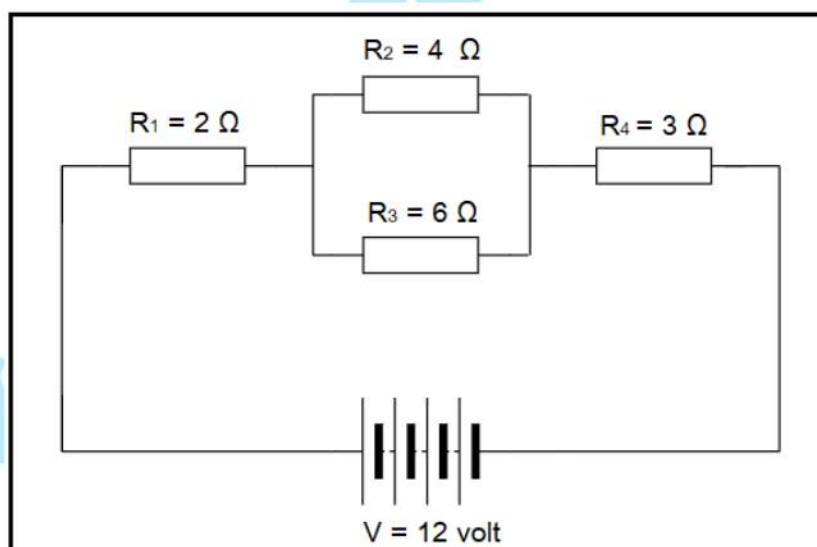


FIGURE 1

- 1.2.1 Refer to the way in which all the resistors are connected to each other and identify the type of circuit connection. (1)
- 1.2.2 Calculate the total resistance of the circuit. (3)
- 1.2.3 Calculate the total current flowing through the circuit. (2)
- 1.2.4 Calculate the total power drawn by the circuit. (2)



12)

-4-

1.3 Conductors come in different shapes and sizes and are made from different materials.

State TWO factors that influence the resistance of a conductor.

(2)

1.4 A transformer has 300 turns on the primary winding and 60 turns on the secondary winding. The transformer has a secondary voltage of 24 V and a primary current of 2 A.

1.4.1 Calculate the primary voltage.

(3)

1.4.2 Calculate the secondary current.

(2)

[20]



QUESTION 2

2.1 Name TWO types of electric shock.

(2)

2.2 Electromagnets are fundamental components in the electrical field, with one of their most common applications being in relays.

Draw a neat and fully labelled circuit diagram of a relay.

(6)

2.3 When wiring a domestic installation, an electrician must always follow the SANS 10142 wiring code to ensure safety and compliance.

Complete the following paragraph by writing only the missing word or words next to the question number (2.3.1–2.3.4) in the ANSWER BOOK.

Any point of a distribution board that has to be reached during normal operation shall not exceed the height of (2.3.1 ...) above the floor (or walking) level. However, the distribution board may be mounted higher if it can be disconnected from the supply by a switch disconnector that is less than (2.3.2 ...) above floor level. Unless a residential distribution board is housed in an enclosure and direct access cannot be obtained by an infant, no part of an indoor distribution board shall be less than (2.3.3 ...) above the floor level and no part of an outdoor distribution board shall be less than (2.3.4 ...) above the ground level.

(4)

2.4 PVC conduit is widely used to protect cables in electrical installations.

List FOUR advantages of using PVC conduit over metal conduit in electrical installations.

(4)

12)

-5-

2.5



Thabo, an electrician, is preparing to wire a new distribution board box for a residential property. Before starting, he gathers all the necessary tools to ensure that the installation process is efficient and compliant.

The old saying 'Use the correct tool for the job' is very important and should always be taken into consideration.

List FOUR tools that Thabo needs to wire the distribution board box correctly.

(4)

2.6

What tool would you use to bend PVC conduit?

(1)

2.7

When selecting and installing cables, it is essential to ensure that they are suitable for the intended application to guarantee safety and efficiency.

List FOUR variables (factors) that should be considered when selecting a cable for an electrical installation.

(4)

[25]

QUESTION 3

3.1

The tools and equipment used by electronics technicians often have special design characteristics.



Give FIVE guidelines when selecting tools and equipment.

(5)

3.2

You are working in the electronics lab, and your supervisor asks you to replace a faulty resistor on a circuit board. You find several resistors in the component drawer, but none of them are labelled with numerical values.

Determine the value and tolerance of the resistors with the following colour bands:

3.2.1 Brown, black, red, silver

3.2.2 Grey, red, red, gold

3.2.3 White, brown, yellow, gold

3.2.4 Blue, grey, green, gold

(4 × 1)

(4)

3.3

As a mechatronics technician, your supervisor assigns you the task of planning an electronic project.

Identify SIX things that should be looked at as part of the planning stage.

(6)

12)

-6-

3.4

Analyse the circuit in FIGURE 2 below.

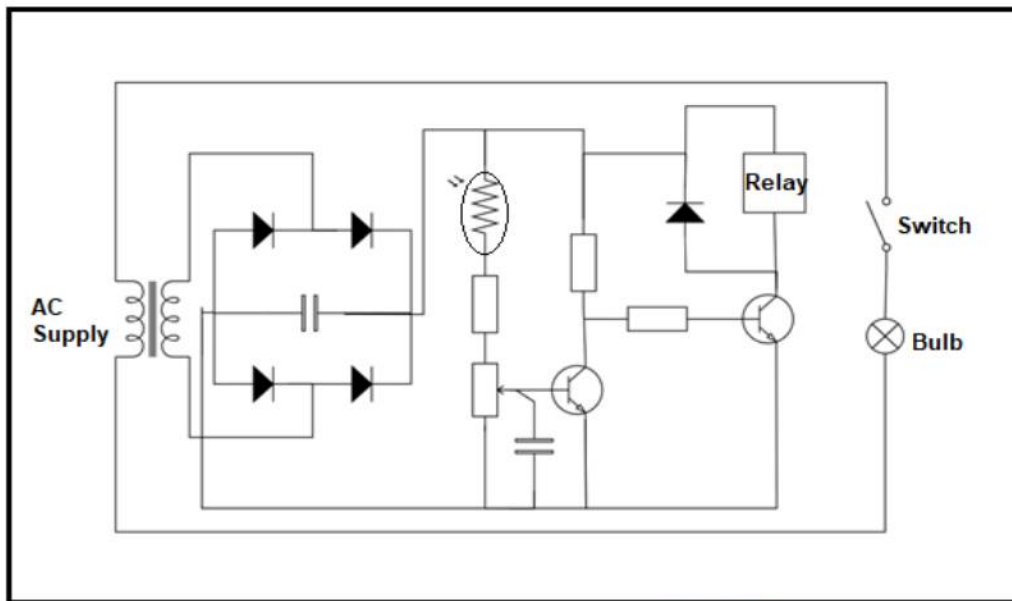


FIGURE 2

Draw up a materials list (exclude the relay, switch and bulb) of components needed to build the day/night switch circuit, as shown in FIGURE 2. Copy and complete the following table in the ANSWER BOOK:

ITEM NUMBER	DESCRIPTION	QUANTITY
1		
2		

(7)

3.5 Resistors can fail due to exceeding their predicted operating life and mechanical strength, or due to vibrations.

Identify THREE ways in which a resistor can fail.

(3)

[25]



QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

- 4.1.1 A loose connection can cause intermittent faults in a circuit.
- 4.1.2 A multimeter can only measure voltage, not current or resistance.
- 4.1.3 All electrical faults can be repaired without turning off the power supply.
- 4.1.4 A ground fault occurs when the current flows through an unintended path to the ground.

12)

-7-

4.1.5 Overheating of wires is a sign of a potential fault in the circuit. (5 × 1) (5)

4.2 Troubleshooting electrical circuits is a fundamental skill for technicians and engineers. It involves identifying and resolving faults that disrupt the normal operation of a system. ★

List FIVE techniques that are useful for troubleshooting. (5)

4.3 Define *electric shock*. (1)

4.4 FIGURE 3 below shows a circuit diagram that is relevant to fault finding.

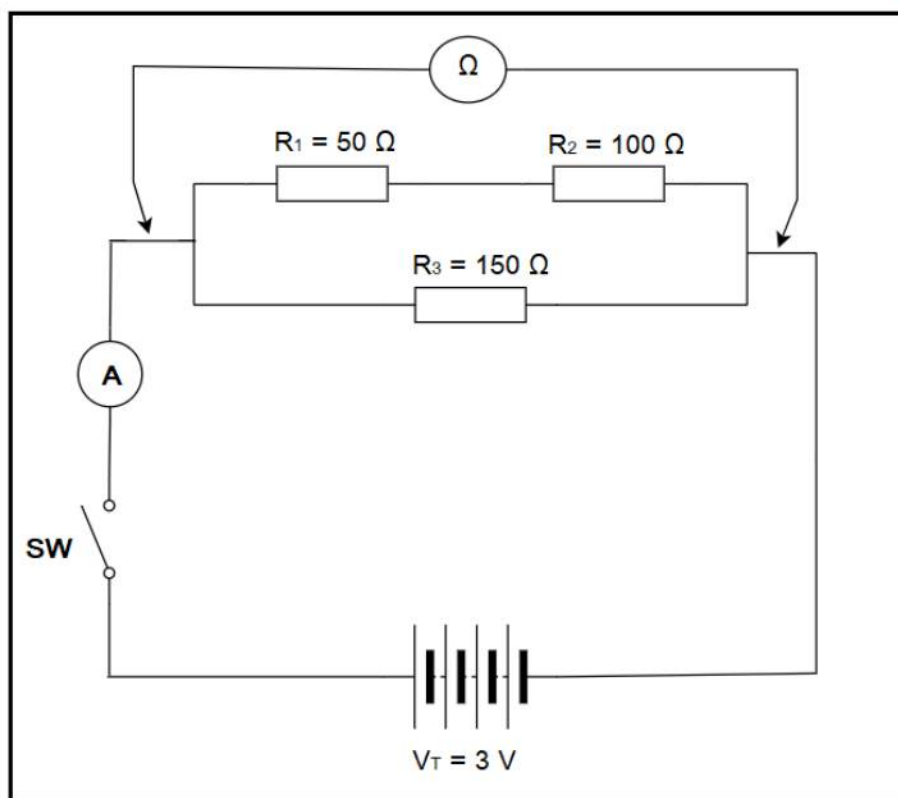


FIGURE 3

4.4.1 When the switch (SW) is in an open position, what will the reading on the ohmmeter be? (3)

4.4.2 After removing the ohmmeter from the circuit and closing the switch (SW), what will the ammeter reading be? (2)

4.4.3 If resistor R_3 is open-circuited, what will the ammeter reading on the circuit be? (2)

4.4.4 If resistor R_1 is short-circuited, what will the reading on the ammeter be? (2)

[20]

12)

-8-

QUESTION 5

5.1 The Compensation for Occupational Injuries and Diseases Act requires employers to report all accidents at work.

List FIVE steps to follow when reporting any accident. (5)

5.2 John is preparing for an upcoming project presentation at work. To ensure that he stays organised, he decides to create a task list.

List FIVE steps you should take in creating a task list. (5)
[10]

TOTAL: 100



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FORMULA SHEET

ELECTROTECHNOLOGY LEVEL 2

$$1. \quad P = V I \cos \phi$$

$$2. \quad Q = V I \sin \phi$$

$$3. \quad S = V I$$

$$4. \quad V = I R$$

$$5. \quad P = V I$$

$$6. \quad X_L = 2\pi f L$$

$$7. \quad X_C = \frac{1}{2\pi f C}$$

$$8. \quad Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$9. \quad V_T = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$10. \quad V_1 = \left(\frac{R_1}{R_t} \right) V_T$$

$$11. \quad V_L = I X_L$$

$$12. \quad I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$$

$$13. \quad \tan \phi = \frac{I_L - I_C}{I_R}$$

$$14. \quad Z = \frac{V}{I}$$

$$15. \quad A = \frac{\pi d^2}{4}$$

$$16. \quad R_T = R_1 + R_2 + R_3 + \dots + R_n$$

$$17. \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$18. \quad C_T = C_1 + C_2 + C_3 + \dots + C_n$$

$$19. \quad \frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

$$20. \quad L_T = L_1 + L_2 + L_3 + \dots + L_n$$

$$21. \quad \frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n}$$

$$22. \quad Q = C V$$

$$23. \quad R_t = R_0(1 + \alpha t)$$

$$24. \quad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$25. \quad I_1 = \left(\frac{R_2}{R_1 + R_2} \right) I_T$$

$$26. \quad V_C = I X_C$$

$$27. \quad R = \frac{p l}{A}$$

$$28. \quad \tan \phi = \frac{X_L - X_C}{R}$$

$$29. \quad E = I^2 R t$$

$$30. \quad n_s = \frac{f}{p}$$

$$31. \quad E = V + I r$$